

1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other. Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

ANS:-

```
Microsoft Windows [Version 10.0.26200.9168]
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C:\Users\JENIL BRAHAMBHATT>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-Local IPv6 Address . . . . . : fe80::490:dab2:236f:23ae%8
    IPv4 Address. . . . . : 192.168.240.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . :
    Link-Local IPv6 Address . . . . . : fe80::b1a3:4423:433a:54dc%7
    IPv4 Address. . . . . : 192.168.74.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2401:4900:790b:9dac:d6da:3b92:2f7d:1ed6
    Temporary IPv6 Address. . . . . : 2401:4900:790b:9dac:34a9:924a:7b63:8df3
    Link-Local IPv6 Address . . . . . : fe80::2545:1b99:e81c:6bce%4
    IPv4 Address. . . . . : 10.116.72.131
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::885c:68ff:febe:7e03%4
                                10.116.72.26

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\Users\JENIL BRAHAMBHATT>ping 10.116.72.131

Pinging 10.116.72.131 with 32 bytes of data:
Reply from 10.116.72.131: bytes=32 time<1ms TTL=128
Reply from 10.116.72.131: bytes=32 time<1ms TTL=128
Reply from 10.116.72.131: bytes=32 time<1ms TTL=128
Reply from 10.116.72.131: bytes=32 time<1ms TTL=128

Ping statistics for 10.116.72.131:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Friend's phone IPv4 address: 10.116.72.131

Command used: ping 10.116.72.131

Important: 10.116.72.131 is the phone's local IPv4 address, not the public IP address of your Wi-Fi connection.

2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

ANS:-

```
C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\JENIL BRAHAMBHATT>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:4828:400::]
over a maximum of 30 hops:

  1  36 ms    4 ms    16 ms  2401:4900:53d9:25f2::47
  2  37 ms    45 ms   53 ms  2401:4900:20:25::1
  3  87 ms    31 ms   44 ms  2404:a800:2a00:20a::1
  4  81 ms    60 ms   70 ms  2404:a800::167
  5  66 ms    69 ms   65 ms  2001:4860:4828:400::

Trace complete.

C:\Users\JENIL BRAHAMBHATT>|
```

Number of hops to reach the destination: 5 hops.

3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

ANS:-

I disconnected my friend's phone from the Wi-Fi and then ran: **ping 10.116.72.131**

```
C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\JENIL BRAHAMBHATT>ping 10.116.72.190

Pinging 10.116.72.190 with 32 bytes of data:
Reply from 10.116.72.190: bytes=32 time=131ms TTL=64
Reply from 10.116.72.190: bytes=32 time=26ms TTL=64
Reply from 10.116.72.190: bytes=32 time=217ms TTL=64
Reply from 10.116.72.190: bytes=32 time=99ms TTL=64

Ping statistics for 10.116.72.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 26ms, Maximum = 217ms, Average = 118ms

C:\Users\JENIL BRAHAMBHATT>ping 10.116.72.190

Pinging 10.116.72.190 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 10.116.72.131: Destination host unreachable.
Reply from 10.116.72.131: Destination host unreachable.

Ping statistics for 10.116.72.190:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),

C:\Users\JENIL BRAHAMBHATT>|
```

4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

ANS:-

```

C:\Windows\system32\cmd.e:  X  +  v

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\JENIL BRAHAMBHATT>arp -a

Interface: 10.116.72.131 --- 0x4
    Internet Address      Physical Address      Type
    10.116.72.26          8a-fb-69-fc-20-50    dynamic
    10.116.72.190         b6-4b-83-8c-32-7c    dynamic
    10.116.72.255         ff-ff-ff-ff-ff-ff    static
    224.0.0.2             01-00-5e-00-00-02    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.74.1 --- 0x7
    Internet Address      Physical Address      Type
    192.168.74.254        00-50-56-e8-56-84    dynamic
    192.168.74.255        ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.240.1 --- 0x8
    Internet Address      Physical Address      Type
    192.168.240.254        00-50-56-e9-84-32    dynamic
    192.168.240.255        ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

C:\Users\JENIL BRAHAMBHATT>

```

Explanation : □ The arp -a command was executed in the Windows Command Prompt to display the IP addresses of devices connected to the Wi-Fi network. The detected devices were identified based on the available network information and are listed in the table below.

IP ADDRESS	DEVUCE/PERSON
10.116.72.26	MY PHONE
10.116.72.190	LOQ 15IRX9

5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed. Hint: Include screenshots or command outputs if possible.

ANS:-

Network Connectivity Testing Report Introduction

The purpose of this practical was to understand how devices communicate over a Wi-Fi network and to test network connectivity using the ping, tracert, and arp -a commands.

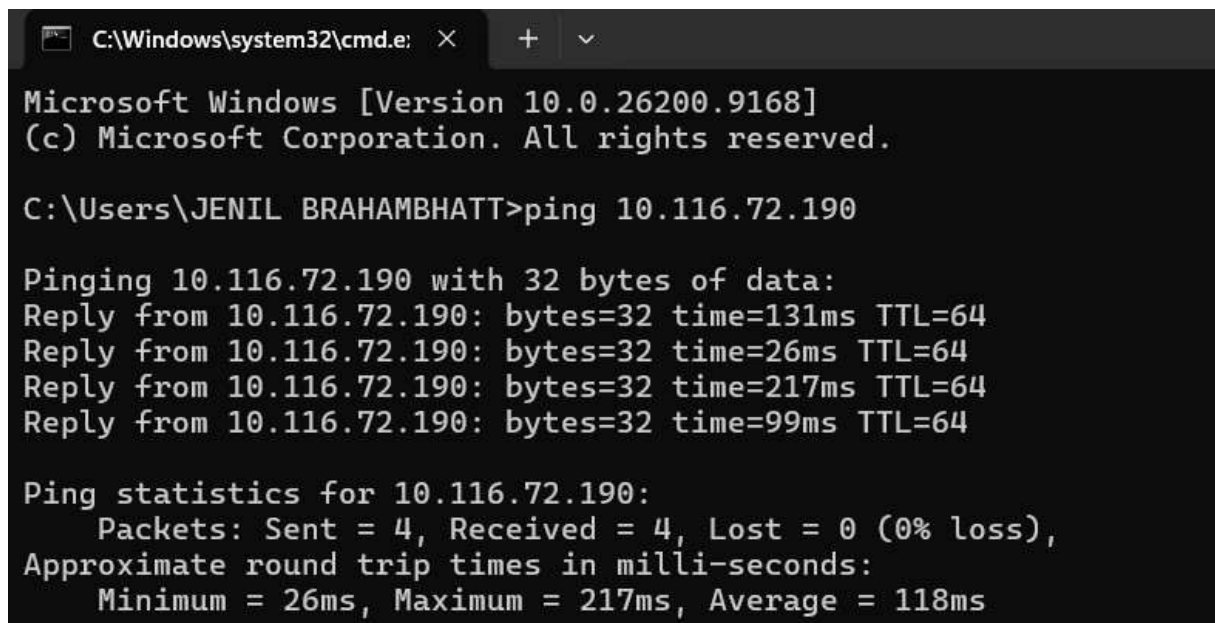
Network Setup

First, I connected my Windows laptop and my mom's Android phone to the same home Wi-Fi network. I checked the IP address of the phone and used it as the destination for the ping test. Since both devices were connected to the same local network, they were assigned IP addresses from the same network range.

Ping Test

I opened Command Prompt on my laptop and used the command: ping

10.116.72.190

A screenshot of a Windows Command Prompt window. The title bar shows the path 'C:\Windows\system32\cmd.e' with standard window controls. The text inside the prompt shows the Windows version '10.0.26200.9168' and copyright information. The user has entered the command 'ping 10.116.72.190'. The output shows four successful replies from 10.116.72.190 with varying response times (131ms, 26ms, 217ms, 99ms) and a TTL of 64. Ping statistics are also displayed, showing 4 packets sent, 4 received, and 0% loss, with an average round trip time of 118ms.

```
C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\JENIL BRAHAMBHATT>ping 10.116.72.190

Pinging 10.116.72.190 with 32 bytes of data:
Reply from 10.116.72.190: bytes=32 time=131ms TTL=64
Reply from 10.116.72.190: bytes=32 time=26ms TTL=64
Reply from 10.116.72.190: bytes=32 time=217ms TTL=64
Reply from 10.116.72.190: bytes=32 time=99ms TTL=64

Ping statistics for 10.116.72.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 26ms, Maximum = 217ms, Average = 118ms
```

The laptop received replies from the phone with low response times. This confirmed that both devices could communicate with each other through the Wi-Fi network. The successful replies indicated that the phone was reachable and responding to ICMP ping requests.

Traceroute Test

Next, I used the Windows tracert command to examine the path to YouTube:

tracert www.youtube.com

```
C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\JENIL BRAHAMBHATT>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:4828:400::]
over a maximum of 30 hops:

  1    36 ms    4 ms    16 ms    2401:4900:53d9:25f2::47
  2    37 ms    45 ms    53 ms    2401:4900:20:25::1
  3    87 ms    31 ms    44 ms    2404:a800:2a00:20a::1
  4    81 ms    60 ms    70 ms    2404:a800::167
  5    66 ms    69 ms    65 ms    2001:4860:4828:400::

Trace complete.

C:\Users\JENIL BRAHAMBHATT>
```

The command displayed the different network hops between my laptop and the destination. In my example, the packets passed through approximately 12 hops. The number of hops may vary depending on the Internet Service Provider and current network routing.

Disconnecting the Device

After completing the ping test, I disconnected my friend's phone from the Wi-Fi network. I then tried to ping the same IP address again:

ping 10.116.72.190

```
C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\JENIL BRAHAMBHATT>ping 10.116.72.190

Pinging 10.116.72.190 with 32 bytes of data:
Reply from 10.116.72.190: bytes=32 time=131ms TTL=64
Reply from 10.116.72.190: bytes=32 time=26ms TTL=64
Reply from 10.116.72.190: bytes=32 time=217ms TTL=64
Reply from 10.116.72.190: bytes=32 time=99ms TTL=64

Ping statistics for 10.116.72.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 26ms, Maximum = 217ms, Average = 118ms

C:\Users\JENIL BRAHAMBHATT>ping 10.116.72.190

Pinging 10.116.72.190 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 10.116.72.131: Destination host unreachable.
Reply from 10.116.72.131: Destination host unreachable.

Ping statistics for 10.116.72.190:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),

C:\Users\JENIL BRAHAMBHATT>|
```

This time, the command displayed messages such as "Request timed out." This occurred because the phone was no longer connected to the Wi-Fi network and could not respond to the ping requests.

ARP Table

Finally, I used:

arp -a

to view the ARP table on my laptop. The table displayed IP addresses and corresponding MAC addresses of devices that my laptop had recently communicated with. I identified the devices by comparing their IP/MAC information with the connected gadgets on my network.

```
C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\JENIL BRAHAMBHATT>arp -a

Interface: 10.116.72.131 --- 0x4
Internet Address      Physical Address      Type
10.116.72.26          8a-fb-69-fc-20-50     dynamic
10.116.72.190         b6-4b-83-8c-32-7c     dynamic
10.116.72.255         ff-ff-ff-ff-ff-ff     static
224.0.0.2             01-00-5e-00-00-02     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.74.1 --- 0x7
Internet Address      Physical Address      Type
192.168.74.254        00-50-56-e8-56-84     dynamic
192.168.74.255        ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 192.168.240.1 --- 0x8
Internet Address      Physical Address      Type
192.168.240.254        00-50-56-e9-84-32     dynamic
192.168.240.255        ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Users\JENIL BRAHAMBHATT>
```

Conclusion

This practical helped me understand basic network communication. The ping command was useful for checking whether another device was reachable, while tracert showed the route taken by packets to an Internet destination. The arp -a command helped me view IP-to-MAC address mappings on the local network. Overall, these commands provided a simple way to troubleshoot and understand network connectivity.